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An R package to Compute Least Squares Sparse Principal Components

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Abstract

This article illustrates the use of the package *spca* how to write a manuscript for the *Journal of Statistical Software* (JSS) using its LaTeX style files. Generally, we ask to follow JSS's style guide and FAQs precisely. Also, it is recommended to keep the LaTeX code as simple as possible, i.e., avoid inclusion of packages/commands that are not necessary. For outlining the typical structure of a JSS article some brief text snippets are employed that have been inspired by **?**, discussing count data regression in R. Editorial comments and instructions are marked by vertical bars.

Keywords: JSS, SPCA, Least squares, projection, uncorrelated, R.

1. Introduction to LS-SPCA

sparse principal components analysis (SPCA) is a variant of principal components analysis (PCA) that aims at computing sparse principal components (sPCs) that are linear combinations of only a few of the features, hence having *sparse* loadings with few nonzero elements.

Most of the existing *conventional* SPCA methods compute sPCs that have maximal variance, using the objective function defined by Hotelling (1933). There exist a plethora of such methods which, as outlined in the *All sparse PCA models are wrong, but some are useful* series of review articles (Camacho, Smilde, Saccenti, and Westerhuis, 2020, 2021, 2025), produce competing solutions, that are often not close to the original principal components (PCs). Merola and Chen (2019) also shows other drawbacks of these methods.

LS-SPCA was first proposed in Merola (2015) and its projection variant in Merola and Chen (2019). This approach aims to compute sPCs that can be used to best approximate the data, in a least–squares (LS) sense, hence adding sparsity constraints to (Pearson 1901) classic PCA's data approximation objective. It should be noticed that, while Pearson's and Hotelling's objectives give the same solutions in PCA, this is not the case when sparsity constraints are introduced(Merola 2015; Merola and Chen 2019), and this is probably the reason why Camacho *et al.* found that all (conventional) SPCA methods are wrong.

The **spca** package computes LS-SPCA solutions with several options and helper functions to visualize and compare the solutions. Rather than a black-box optimizer type function, the package provides an intuitive UI with transparent options and goodness–of–fit metrics.

This article will be used as extended vignettes in the package, so will illustrate several of the available options and helpers, useful to conduct a thorough SPCA. In the next Section we briefly introduce the theory behind LS-SPCA. In Section **??** we illustrate the package with examples.

1.1. LS-SPCA estimation

The material in this section is taken from Merola (2015) and Merola and Chen (2019) where proofs and further details can be found. Let's establish notation first. X denotes the $n \times p$ data matrix containing n cases on p features centered to mean zero. $S \propto X^\intercal X$ denotes the covariance matrix. The eigenvectors of S, $v_1, \ldots, v_p$ ($\|v_j\| = 1,\ j = 1, \ldots, p$ (the PCs loadings) are ordered by decreasing corresponding eigenvalues, $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_p$. The same convention is used for all eigenvectors (including generalized ones).

The sparse loadings are the vectors $a_j = (a_{i1}, \ldots, a_r, r \leq p$ with unit norm $a_j^\intercal a_j = 1$, each of which has only $c_j \leq p$ non-zero elements. The quantity c_j is called the *cardinality* of a_j. The sPCs are then equal to $t_j = Xa_j$.

A dot above a symbol denote the operation of retaining the elements of a vector or the columns of a matrix corresponding to a set of indices, ind_j. In R syntax, we write $\dot{\mathbf{a}}_j = a_j[ind_j]$ to indicate the vector containing only the c_j non-zero loadings in j, so that $t_j = \dot{X}_j \dot{a}_j$.

There exist there variants of LS-SPCA, optimizing two different objectives LS-SPCA solutions are derived using two different objectives.

Objective: maximal variance explained

This objective is obtained by adding sparsity constraints to the standard least squares PCA objective of maximizing the variance explained (VEXP).

$$\min_{a_j} \|X - t_j b^\intercal\|^2, \quad \text{subject to} \quad \text{card}(a_j) < p, \quad a_j^\intercal S a_i = 0,\ i < j; j = 1, \ldots, r. \tag{1}$$

Uncorrelated LS-SPCA (uSPCA). The optimal sparse loadings for objective 1, $\dot{\mathbf{a}}_j$, are the generalized eigenvectors corresponding to the largest eigenvalue, λ_j, solution to

$$C_j \dot{\mathbf{S}}^\intercal \dot{\mathbf{S}} \dot{\mathbf{a}}_j = \lambda_j \dot{X}_j^\intercal \dot{X}_j \dot{\mathbf{a}}_j, \tag{2}$$

where $\dot{\mathbf{S}} = S[ind_j]$ and the matrices C_j enforce the orthogonality constraints ($C_1 = I$).

The orthogonality constraints require that the cardinality of the solution loadings satisfy $c_j \geq j$. Furthermore, the solutions can be computationally demanding and unstable.

Correlated LS-SPCA (cSPCA). Approximated but simpler to compute solutions can be obtained from the residual matrices orthogonal to the previously computed sPCs. Let

$$Q_j = \left(I_p - \frac{t_{j-1}t_{j-1}^\intercal}{t_{j-1}^\intercal t_{j-1}}\right) Q_{j-1}, \quad j = 2, 3, \ldots, r, \tag{3}$$

with $Q_1 = X$. In SPCA literature the residual matrices Q_j are called *deflated* matrices. Note that the use of the deflated matrices is a direct consequence of the Courant–Fischer min–max theorem and is typically used in the power method for computing the eigen-decomposition. Then, the cSPCA solution loadings are obtained as the generalized eigenvectors corresponding to the largest eigenvalue:

$$Q_j Q_j^\intercal \dot{X}_j \dot{\mathbf{a}}_j = \mu_j \dot{X}_j^\intercal \dot{X}_j \dot{\mathbf{a}}_j. \tag{4}$$

The first cSPCA solution is equal to the first uSPCA solution but, in general, the following solutions are not equal to the corresponding uSPCA solutions and, typically show very mild correlation with one another, with increasing values for sPCs of higher order. However, their cardinality is not constrained by the uncorrelatedness requirements and, in most cases, it is lower than their order.

Maximal correlation with the PCs (pSPCA)

The sPCs obtained by maximizing the variance explained are usually highly correlated with the corresponding PCs. However, as the order of the components increases, these correlations tend to decrease.

The pSPCA objective function requires that the sPCs best approximate the PCs. That is

$$\max_{a_j} |\mathrm{cor}(t_j, PC_j)|, \quad \text{subject to}\, \mathrm{card}(a_j) < p. \tag{5}$$

It is well known that the optimal solution to this objective is linear regression. Since staring with the first, the sPCs explain less variance than the corresponding PCs, using the first PC of the Q_j matrices, instead of the PCs, allows to recover some of this unexplained variance.

Therefore, in pSPCA, the sparse loadings are computed as the coefficients, standardized to unit norm, of the regression of the residual PCs onto subsets if the x-variables. That is

$$\dot{\mathbf{a}}_j \propto (\dot{X}_j^\intercal \dot{X}_j)^{-1} \dot{X}_j^\intercal z_j, \tag{6}$$

where z_j denotes the j-th residual PC.

In general, the pSPCA loadings are close to the cSPCA ones but show higher mutual correlation, still of small magnitude. This approach ensures best approximation sPC to PC and is computationally simpler. Furthermore, pSPCA provide a very efficient approach to variable selection.

1.2. Variable selection

Variable selection is notoriously a computationally demanding (NP-hard) task that requires greedy algorithms for its solution.

For uSPCA and cSPCA we implemented *intensive* forward selection algorithms based on the actual variable explained that terminate after a minimal proportion of the cumulative variance explained (CVEXP) of the corresponding number of PCs is reached. These are computationally expensive, especially for large matrices, because they require to repeatedly compute generalized eigen-decompositions.

We also implemented efficient regression based standard forward, backward, and forward stepwise variable selection algorithms. These algorithms always use the computationally cheap R^2 to select or eliminate variables. For maximum sPC-PC correlation objective the current R^2 should be used as stopping rule. For maximum VEXP objective the current CVEXP should be used. In this case, at every search step, the current CVEXP is computed to evaluate the stopping rule, so they can still be computationally expensive. Forward selection is advised when p is large. Different types of variable selection are chosen via the argument `var_selection = c("fwd", "bkw", "step")` and the stop criterion via the argument `stop_criterion = c("r2", "cvexp")`. The variance explained based intensive selection is selected by setting `intensive = TRUE`.

2. Computational details

in **spca**, linear algebra and matrix is handled by the C++ back-end. This is used also to prevent deep copies typically made by R functions. The exported functions are called from R wrappers.

The main function `spca()` is used to compute LSSPCA solutions of all types with all options. Different back-ends manage "fat" matrices ($n < p$) and "tall" matrices. The tall back-end operates on the covariance matrix S, while the fat back-end operates on the row-space of the data matrix, using a *reverse-SVD* approach to compute eigen-decompositions and dedicated variable selection algorithms to compute LS-SPCA solutions. Which back-end is used is decided automatically, based on whether the data matrix aspect-ratio is less of one or not, if `fat_matrix = NULL`, or it can forced by setting `fat_matrix = TRUE/FALSE`.

The major expedient that we use is to do the potentially demanding $O(np^2)$ multiplication to compute the covariance matrix $S = X^{\mathsf{T}}X$ and the $O(p^3)$ multiplication SS only once at the beginning. Then deflations are done with cheap rank-1 updates instead of recomputing tthe natrices as products of the deflated data matrix. The computational savings can be substantial when p is large, as they reduce the cost from , roughly, $O(p^3)$ to $O(p^2)$.

Since, apart from the initial eigen-decomposition, only the leading eigen-pair of a matrix is needed, these can optionally be computed with the power method. The power method option can be triggered by setting `PML = TRUE` for loading computation and `PMVS = TRUE` for intensive variable selection computation. Using the power method reduces the computational cost for a $p \times p$ matrix from approximately $O(p^3)$ to approximately $O(Kp^2)$, where K is the number of iterations required for convergence. The convergence can be controlled with the `maxiterPM...` and `epsPM...` for each set of computations.

Regression based variable selection algorithm is simplified by considering that the regressed variable is an eigen-vector of the regressors and uses rank-1 updates for computing current values of selected stopping criterion.

Similar considerations are valid for the fat back-end engine.

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